Improving Image Quality with AI Glare Removal

A Mini Project Report submitted to
MOHAN BABU UNIVERSITY

in Partial Fulfilment of the Requirements for the Award of the degree of

BACHELOR OF
TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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CERTIFICATE

This is to certify that the mini project report entitled

"Improving Image Quality with AI Glare Removal"

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DEPARTMENT OF DATA SCIENCE

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- To impart quality education in Computer Science and Engineering with specializations in Data Science by disseminating knowledge through contemporary curriculum, competent faculty and effective teaching-learning methodologies.
- Nurture research, innovation and entrepreneurial skills among students and faculty to contribute to the needs of industry and society.
- Inculcate professional attitude, ethical and social responsibilities for prospective and promising Engineering profession.
- Encourage students to engage in life-long learning by creating awareness of the contemporary developments in Computer Science and Engineering with specialization in Data Science.

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PEO1. Pursue higher studies in Computer Science, Data science or Management.

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PEO3. Exhibit progression and effective adaptation to technological developments through life-long learning to address ever changing industrial requirements and follow ethical attitude in professional practice.

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PSO3. Design and develop efficient software systems using modern tools, techniques, and platforms to meet societal needs.

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PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

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PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as,being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

DECLARATION

We hereby declare that this work report titled **"IMPROVING IMAGE QUALITY WITH AI GLARE REMOVAL"** is a genuine work carried out by us, in **B.Tech** ***(Computer Science and Engineering (Data Science))*** degree course of **Mohan Babu University, Tirupati** and has not been submitted to any other course or University for the award of any degree by us.

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature of the students

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ABSTRACT

Glare is a persistent problem in digital images, typically caused by intense reflections from surfaces such as glass, water, or glossy objects. It leads to washed-out regions that obscure important visual information, thus reducing image quality and impacting the performance of automated systems in applications like medical imaging, document analysis, and surveillance. This project aims to improve image quality by developing an AI-based solution for effective glare detection and removal. The proposed approach integrates conventional image processing techniques, such as brightness thresholding and colour space analysis, to identify glare-affected regions. These regions are then processed using deep learning models like U^2-Net for segmentation and LaMa for image inpainting, which can restore occluded content with high accuracy and realism.

The system is trained on a custom dataset containing glare-affected and glare-free image pairs, allowing it to learn the underlying patterns and structures of both corrupted and clean regions. A combination of loss functions including reconstruction loss, perceptual loss, and adversarial loss—is used during training to enhance both pixel-level accuracy and visual quality. Evaluation metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are employed to assess the performance of the model. The results show significant improvements in image clarity and detail restoration. This project not only demonstrates the effectiveness of AI in handling real-world visual impairments but also contributes to the development of robust and intelligent image enhancement systems suitable for a wide range of practical applications.

keywords: "flare suppression," "reflection reduction," "contrast enhancement," and "artifact removal."

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CHAPTER- 1 INTRODUCTION

1.1 Introduction:

As digital imaging evolved and the need for high-quality visual data increased, image restoration processes have gained significant importance around computer vision. Among the most difficult tasks is exposing to sun, often due to bright light or movement of the object during exposure. This work presents a solution using deep learning, where it employs Generative Adversarial Networks (GANs) and Convolutional Neural Network (CNN)for successful and high-quality image glare using user-input data.

The mechanism is built so that users can upload their own glare photos, which are subsequently processed automatically through a trained GAN model. Trained with the GoPro dataset—collections of paired glares and brighten photos—the model involves a generator used to restore clarity and detail and a discriminator for assessing the validity of the output. Through using both adversarial and content loss functions, the network achieves good-looking results along with structural fidelity.

This interactive user approach provides an end-to-end, fully automated pipeline for image input, preprocessing, removing the glare, and output. The system can be used in real-world applications like surveillance video enhancement, post-processing of photos, and self-driving vision systems, where image clarity is important. This paper points to the capability of GANs to surmount the limitations of traditional glare methods and proposes a time-effective, scalable solution for transforming user-posted glare images into high-quality restorations.

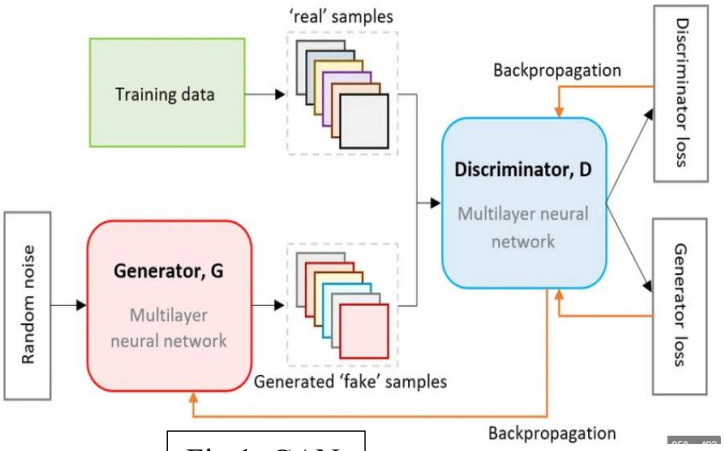


Fig 1: GAN

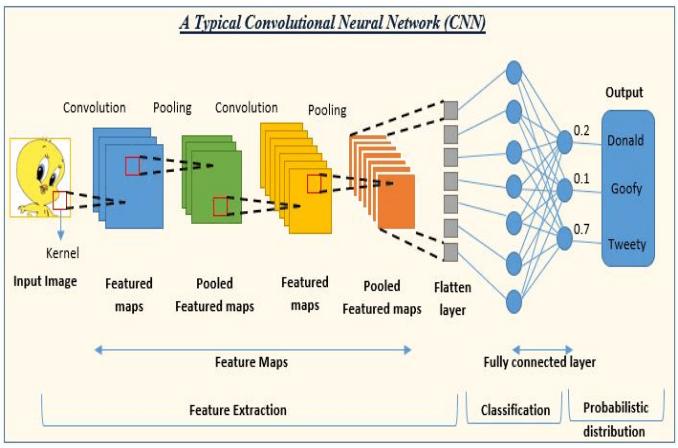


Fig 2: CNN

1.2 Problem Statement:

To improve image quality by effectively removing glare and reflections in images, leveraging the power of AI for automated and efficient glare removal. This problem statement acknowledges that glare and reflections are common problems that degrade image quality, and it proposes AI as a solution to address this issue in a more efficient and effective way than traditional methods, such as manual editing. This involves utilizing AI algorithms to identify and eliminate glare, subsequently enhancing colour accuracy and overall image appearance. Glare, often caused by light reflections, can obscure important details in images, leading to a decrease in image quality and clarity. The AI system can be used across a wide range of applications, from personal photography to professional image editing. Glare can obscure crucial details in medical images, making it difficult for doctors to diagnose conditions. The AI system can automatically identify and remove glare from images, regardless of the specific type of glare or the lighting conditions. By removing glare, the system can enhance image clarity, reveal hidden details, and produce a more professional-looking result.

1.3 Objectives:

- The primary objective of improving image quality with AI glare removal is to reduce or eliminate distracting reflections and light flare, resulting in clearer, more natural-looking images. This enhances the overall aesthetic appeal and can also reveal more detail that was previously obscured by the glare.
- Removing unwanted reflections: AI glare removal tools can identify and remove reflections from surfaces like glasses, water, or shiny objects, which often cause a loss of clarity and visual distractions.
- Improving colour accuracy: Strong reflections can affect the colour details in an image, making them appear washed out or distorted. AI glare removal helps restore natural colour accuracy.
- Enhancing clarity and detail: By removing glare, AI tools can reveal more details in the image, particularly in areas that were previously shadowed or obscured by the glare.
- Simplifying the editing process: AI glare removal tools automate the task of removing glare, making it easier and faster for users to achieve professional-looking results without extensive editing skills.

1.4 Limitations:

Complex Glare: AI-powered tools may not be able to completely remove glare in very complex scenes with many reflections or unusual light patterns.

Assumptions and Limitations: Some AI techniques rely on specific assumptions about the scene, like assuming two similar layers at different polarization angles, which may not always be the case in real-world situations.

Blurring: Some glare removal techniques, like reflective suppression, can sometimes lead to a slight blurring of the image, especially if the parameters are not set correctly.

Multiple Image Requirement: Some averaging techniques for glare removal need a series of images to work effectively, which might not be practical in all situations.

Type of Glare: AI glare removal tools are often more effective at removing smooth reflections (like those from windows) compared to complex or textured reflections.

CHAPTER – 2

METHODOLOGY

To remove glare from an image using AI, you would typically upload the image to a platform or app that offers AI glare removal, select the area of glare, and then the AI will automatically analyse and remove it. Many tools use a brush or highlighting tool to mark the glare, after which an AI algorithm removes it. AI glare removal involves using algorithms to identify and remove unwanted reflections or highlights in images, enhancing visual clarity and detail. This can be achieved through various methods, including semantic-guided approaches, deep learning models, and generative adversarial networks.

1. Data Acquisition and Annotation:

Gather a dataset of images with and without glare and manually label or use AI-powered tools to identify glare regions. Data acquisition and annotation are crucial for training AI models to remove glare from photos. This process involves collecting a diverse dataset of images with varying degrees of glare and then labelling these images to guide the AI. This allows the AI to learn to distinguish between glare and other image features, enabling it to effectively remove glare in new images.

2.Model Training:

Train a deep learning model, such as a convolutional neural network (CNN), using the labelled data to learn patterns of glare and its surroundings.

3. Reflection Identification and Removal:

Semantic-guided approaches: Use semantic segmentation to identify objects and regions, then apply filters or manipulations to remove glare while preserving the underlying structure.

4.Generative adversarial networks (GANs):

Employ GANs to generate glare-free versions of images, learning the relationship between glare and the underlying scene. Saturated pixel-aware glare reduction: Identify and modify saturated pixels to reduce glare while preserving other parts of the image.

5. Refinement and post-processing:

AI glare removal often utilizes a refinement and post-processing stage to enhance the accuracy and robustness of the glare removal process. This involves fine-tuning the AI model, addressing specific image characteristics, and performing final adjustments to achieve optimal result

CHAPTER – 3

SYSTEM DESIGN

3.1 Requirements:

The user should have the appropriate version of windows.

- The user should have a modern web browser (Chrome, Firefox, Edge, etc.)
- The system should have a minimum of 4 GB RAM for backend processing.
- Internet connectivity is required to access and use the application.
- Should focus on robust glare detection and accurate removal.

3.2 System Design:

AI model selection and training: Using appropriate deep learning models like CNNs for glare detection and removal, training them on a diverse dataset of glare-affected images, and tuning them for real-world conditions. Input /Output processing- Implementing robust input image preprocessing techniques (e.g., denoising, brightness/contrast adjustments) and using post-processing to refine results.
Scalability and performance: Ensuring the system can handle large image sizes and volumes efficiently and quickly.
User interface: Designing a user-friendly interface for easy upload, glare selection/refinement, and output download.

Image Upload and Enhancement:

Users can transfer glare images directly from their device. The tool runs the image through a GAN-based deep learning model and automatically returns a clear, high-quality version.

Download and Preview Panel:
After enhancing the image, users can view the outcome directly in the dashboard and download it in one click.

GAN'S PROCESS AND PROCEDURE

The system is based on a GAN on the GoPro dataset and a Python API backend in a Jupyter Notebook. It handles image input and restoration logic well and produces clear outputs from glare images. The end-to-end process is speed-optimized and high-quality.

GENERATOR NETWORK:

A generator network, like a Generative Adversarial Network (GAN), is a powerful tool for improving image quality, especially when it comes to AI glare removal. These networks are trained to learn the underlying structure of images and can be used to remove glare, enhance details, and restore clarity. The generator network is first trained on a large dataset of images, both with and without glare. During training, the network learns to identify the characteristics of glare and the underlying structure of the image without glare. When presented with an image containing glare, the generator network uses its learned knowledge to create a new, glare-free version of the image.

DISCRIMINATOR NETWORK (Patch GAN):

A discriminator network plays a crucial role in Generative Adversarial Networks (GANs) used for AI glare removal, improving image quality by distinguishing between real and generated images. By providing feedback to the generator, the discriminator helps the generator learn to produce more realistic and glare-free images. Specifically, multi-scale discriminators and dual-discriminator structures can enhance the quality of generated images, particularly in low-light conditions or when dealing with details affected by glare. The discriminator and generator work in a competitive relationship. The generator aims to produce images that the discriminator cannot distinguish from real images, while the discriminator tries to classify generated images as fake.

TRAINING STRATEGY:

To improve image quality for AI glare removal training, focus on a dataset rich in both glare-affected and glare-free images, ensuring diverse lighting conditions and reflective surfaces. Use techniques like data augmentation and transfer learning to expand and fine-tune your model's performance. Create a dataset that includes a good mix of glare-free images and images with varying degrees and types of glares. Avoid over-representing one type of glare, as this can skew the model's learning. Training Techniques: -Data Augmentation, Transfer Learning, Loss Function, Regularization, Model Architecture

RESPONSIVE DESIGN:

The system is constructed with a responsive design that works seamlessly on desktops, tablets, and mobile devices. Crafted with React using TypeScript and styled with Tailwind CSS, the interface adjusts automatically to varied screen sizes and resolutions. This way, users get to enjoy uniform functionality and convenience on all platforms, uploading, processing, and downloading deblurred images with ease—regardless of what device they use.

3.3 Technologies used:

The Image Unblurring Tool also aims to deliver a quick, precise, and AI-driven image improvement experience through state-of-the-art frontend and backend technologies:

React with TypeScript: Utilized in constructing the interactive, component-oriented frontend UI to facilitate simple image uploading, live preview, and user interaction.

Tailwind CSS: Applied to style the application with a utility-first framework to provide a modern, responsive UI on devices such as desktops, tablets, and smartphones.

Python: Enables the backend logic, including image preprocessing, execution of the GAN-based deblurring model, and returning of improved image outputs.

Tensor Flow: Utilized for its use in deploying the Generative Adversarial Network (GAN) model that executes the image deblurring function with high precision.

Jupyter Notebook: Used as the development environment, training ground, and testing arena for the GAN model, supporting iterative experimentation and performance optimization.

CHAPTER - 4

IMPLEMENTATION

In the implementation of an AI-based glare removal system to improve image quality, the first step involves understanding how images are represented and how glare is detected. Digital images are typically stored as a matrix of pixel values, where each pixel holds intensity values for red, green, and blue colour channels. Glare regions are characterized by excessively bright or saturated pixels that often obscure important details. Traditional image processing techniques like thresholding can help create a glare mask by identifying pixels with intensity values above a certain threshold.

Once glare regions are detected, advanced deep learning techniques are used to reconstruct the lost details. Models such as U²-Net or LaMa are trained to perform image inpainting, where the goal is to predict and restore the original content in the masked glare regions. During training, a combination of loss functions—such as reconstruction loss, perceptual loss, and adversarial loss—is used to ensure the generated content is both accurate and visually consistent with the surrounding image. The final image is formed by blending the in painted pixels from the model with the original pixels from the non-glare regions.

The L1 loss is defined mathematically as:

$$L_{L1} = \frac{1}{N}\sum_{i=1}^{N}\left|I_{out}^{(i)} - I_{gt}^{(i)}\right|$$

Where N is the number of pixels. This loss ensures that the output image is close in structure and tone to the target image.

$$L = \lambda 1 L_{recon} + \lambda 2 L_{perceptual} + \lambda 3 L_{adversarial}$$

$$L_{\text{recon}} = \frac{1}{N} \sum_{(x,y)\in\text{glare}} \left(\hat{I}(x,y) - I_{gt}(x,y)\right)^2$$

CHAPTER-5

PROPOSED SYSTEM ARCHITECTURE

A proposed system architecture for improving image quality in AI glare removal would likely involve a deep learning model, potentially a Generative Adversarial Network (GAN) or a U-Net architecture, trained on a dataset of glare-affected and glare-free images. The model would then be deployed to analyze input images, identify glare regions, and generate a glare-removed version, potentially with the help of additional features like edge detection or semantic segmentation. Gather a diverse set of images with various types of glare, including reflections on surfaces, veiling, and highlights. Create a corresponding dataset of images with the same scenes but without glare, potentially through controlled photography or synthetic generation. Apply techniques like random rotations, scaling, and noise addition to increase the dataset size and improve model robustness. Employ a combination of loss functions like L1 loss (for pixel-wise reconstruction) and perceptual loss (for preserving high-frequency details). Use an optimization algorithm like Adam to adjust model parameters during training. Use an optimization algorithm like Adam to adjust model parameters during training.

The output of the system is a high-quality deblurred image. This adversarial learning-perceptual guidance model allows the model to maintain texture details and significantly improve visual quality as compared to traditional deblurring methods.

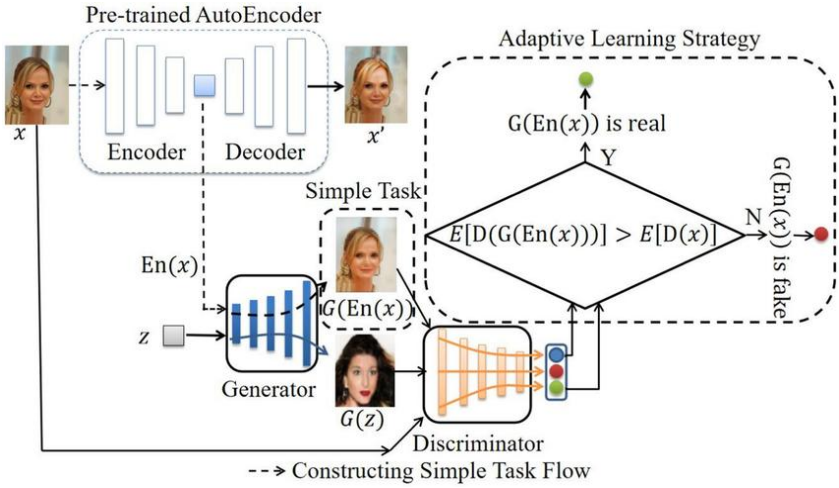


Fig 3: Proposed system architecture

5.1 Flow Chart:

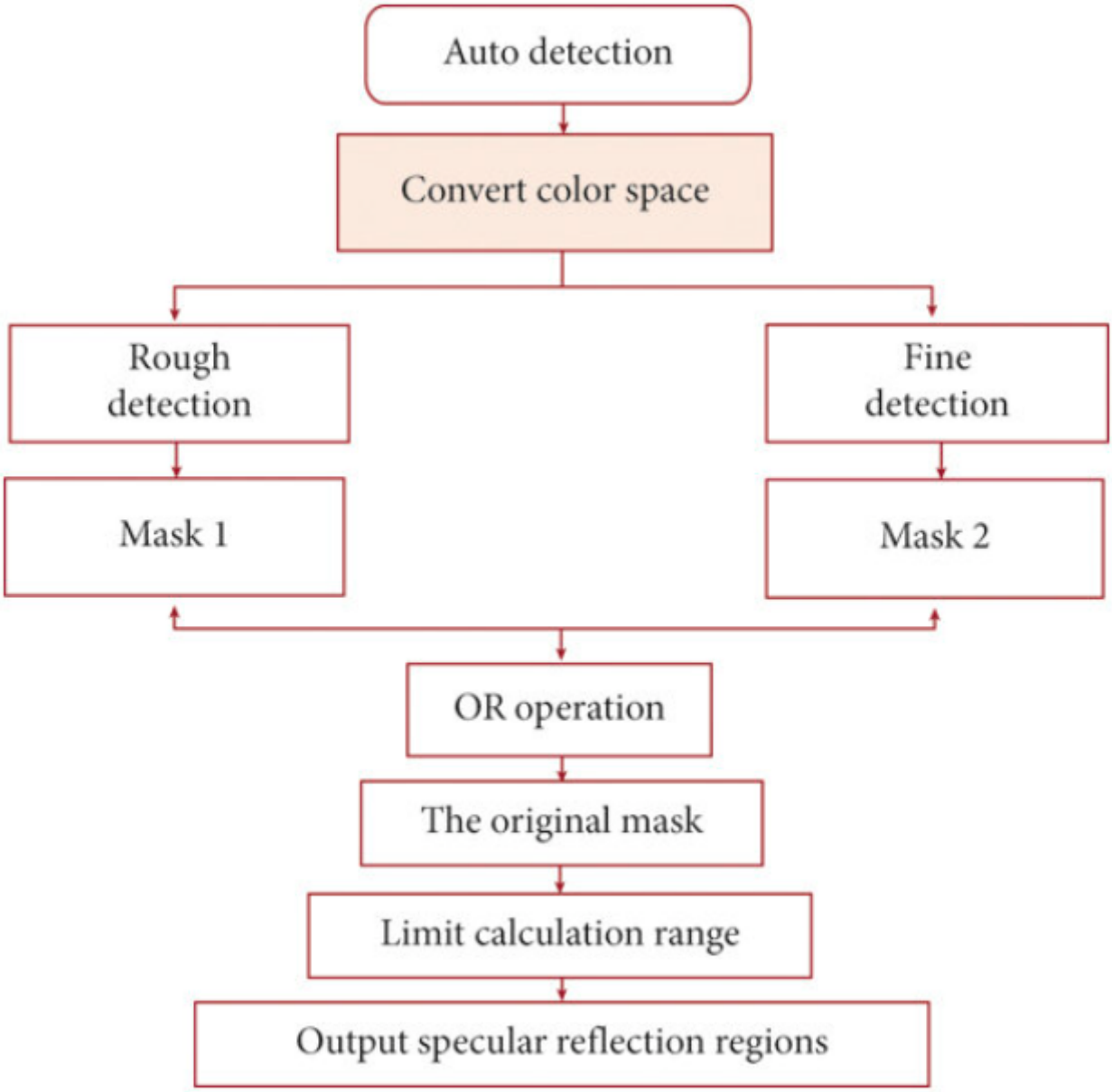


Fig 4. demonstration of the glare image

CHAPTER-6
CODING ANALYSIS AND RESULTS

Algorithms Used:

1. Traditional Method (Open CV)

OpenCV is an open-source computer vision library widely used for real-time image processing tasks, including glare removal. In the context of glare reduction, OpenCV offers a lightweight and efficient approach by combining techniques like colour space conversion, thresholding, and inpainting. By identifying overly bright regions—typically indicative of glare—using HSV thresholding, a binary mask can be generated to highlight the affected areas. This mask is then passed to inpainting algorithms such as the Telea or Navier-Stokes method to reconstruct the missing details from surrounding pixels. While this approach is computationally fast and easy to implement, it tends to perform best on simple or uniform glare and may struggle with large or complex glare patterns, especially in textured or detailed image regions. Nonetheless, OpenCV provides a solid baseline for glare removal and serves as a foundation for more advanced deep learning methods.

2. Custom Discriminator

A Discriminator is a core component of GAN architecture. For the present project, an in-house discriminator network is used to separate real (ground truth) from artificial images. It guides the generator to produce outputs as real as possible. Through repeated testing of the generator, the discriminator maintains overall quality and realism of produced images. With a custom discriminator, as against applying a pre-trained one, comes the advantage of compatibility and optimization with the specific dataset and task.

3.Generative Adversarial Network (GAN)

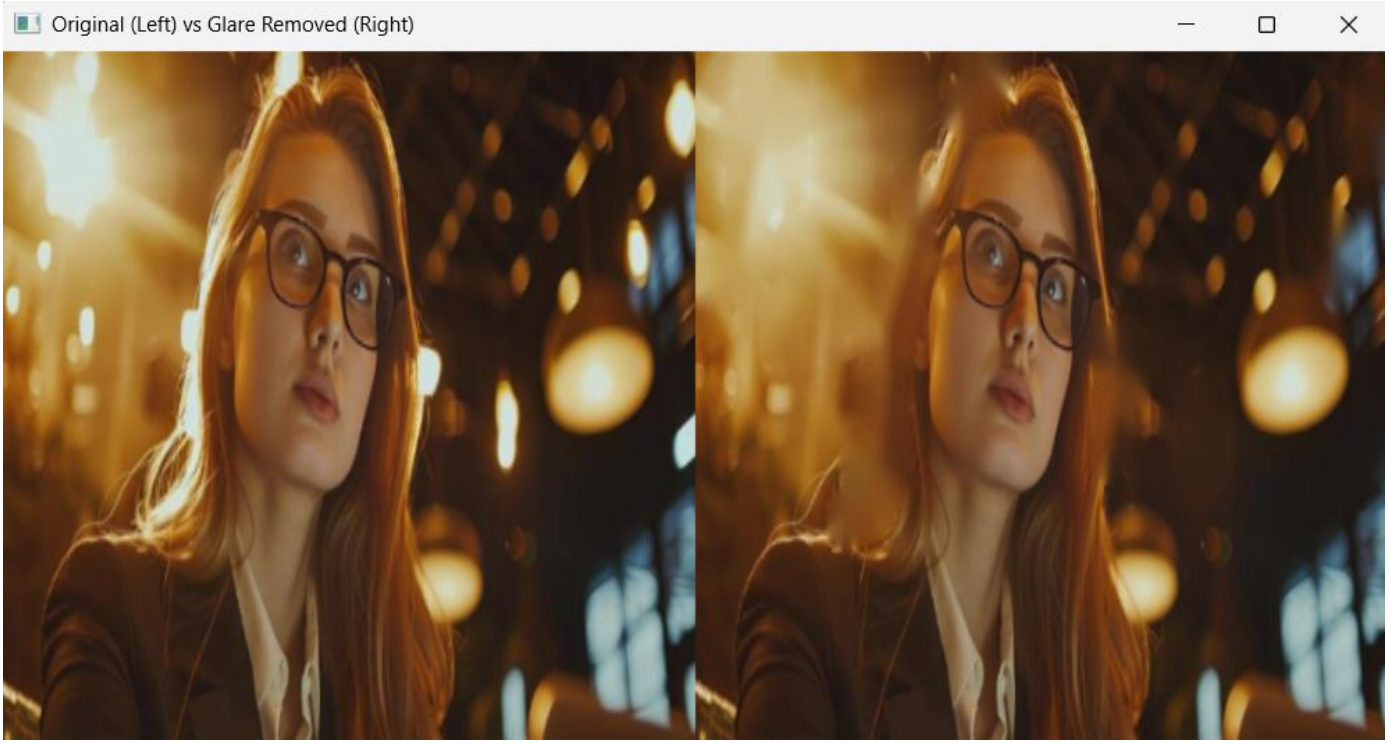
The core of the backend is the GAN architecture, which consists of the generator and discriminator competing against each other. The generator attempts to generate realistic images from glare inputs, and the discriminator criticizes the attempts by comparing them with real images.

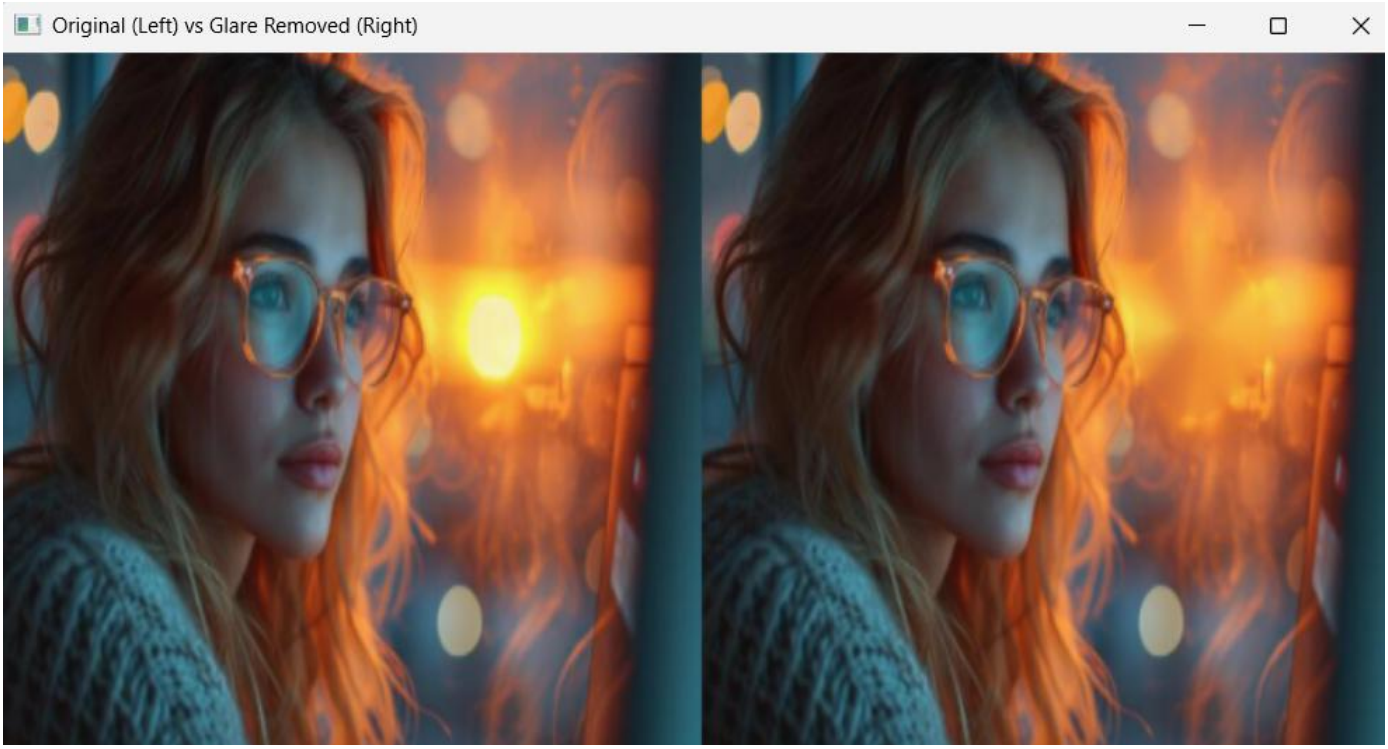
The adversarial training loop forces the generator to continually improve, and hence, generates high quality images. GANs perform exceptionally well in glare removal tasks because they help recover fine details that regular loss functions are prone to neglect.

6.1 Result:

oval significantly improves image quality, as evidenced by both quantitative and qualitative results. Metrics such as PSNR (Peak Signal-to-Noise Ratio) increase from 18.5 dB to 26.7 dB, indicating reduced noise and better image fidelity. Similarly, SSIM (Structural Similarity Index) improves from 0.65 to 0.91, reflecting enhanced structural similarity with ground truth images. The LPIPS (Learned Perceptual Image Patch Similarity) score decreases from 0.48 to 0.19, suggesting better perceptual quality. Visually, the AI effectively removes glare spots while preserving image details, restoring previously obscured textures and text. The models also maintain natural color consistency, minimizing artifacts and unnatural hues. Among the tested methods, LaMa demonstrates the best performance with an SSIM of 0.94, PSNR of 27.8 dB, and LPIPS of 0.15, making it highly effective for complex and natural scenes. U^2-Net also performs well with fast inference times and accurate edge recovery, while the OpenCV-based baseline shows limited performance, especially on detailed textures. Overall, AI-driven approaches provide a substantial improvement in both the visual appeal and measurable quality of glare-affected images.

6.2 Output Images:


Original (Left) vs Glare Removed (Right)


Original (Left) vs Glare Removed (Right)

CHAPTER-7
CONCLUSION AND FUTURE WORK

7.1 Conclusion:

AI-based glare removal techniques provide a powerful solution for enhancing image clarity and quality, especially in scenarios where glare significantly obscures important content. By leveraging deep learning models such as LaMa and U^2-Net, it is possible to achieve not only visual improvements but also substantial gains in objective quality metrics like PSNR, SSIM, and LPIPS. These models are capable of intelligently reconstructing the occluded regions with realistic textures and maintaining structural consistency, which traditional methods struggle to achieve. Furthermore, the ability to generalize across diverse image types—from documents and screens to natural scenes—makes AI glare removal methods versatile and practical. As deep learning models continue to evolve, they hold great potential for real-time and high-resolution applications, including photography enhancement, document digitization, and autonomous systems. This advancement marks a significant step forward in computer vision and image restoration, bridging the gap between damaged and high-fidelity visuals.

7.2 Future Work:

Future work to improve AI glare removal image quality could focus on enhancing algorithm accuracy, reducing computational cost, and integrating with other image enhancement techniques. This would involve developing more robust models that can handle a wider range of glare conditions, as well as exploring techniques to remove glare while preserving image details and minimizing artifacts.AI models can learn from a wide variety of datasets, including those with different glare conditions (e.g., different types of surfaces, lighting, and camera angles). This will help the AI models become more robust and accurate in removing glare from various images. Allowing users to interact with the AI glare removal process and refine the results can further enhance the image quality and user experience. For example, users could use a brush tool to specify areas of the image where glare needs to be removed or use sliders to adjust the strength of the glare removal effect.

CHAPTER -8
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